Enhancing LLM Robustness Against Adversarial Attacks

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Dissertation submitted in the partial fulfilment of the requirements for the degree of

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CERTIFICATE

Certified that this report titled **"ENHANCING LLM ROBUSTNESS AGAINST ADVERSARIAL ATTACKS"** for the Project Work 1 (19Z720) is a bonafide work **of Arjun Mahajan (22z309), Dhanushkumar M (22z317), Jeyanth V P (22z331), Santhosh G (22z355), and Tharun D (23z466)** who have carried out the work under my supervision for the partial requirements for the award of the degree of Bachelor of Engineering in Computer Science and Engineering. Certified further that to the best of my knowledge and belief, the work reported herein does not form part of any other thesis or dissertation based on which a degree or an award was conferred on an earlier occasion.

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SYNOPSIS

This project evaluates the safety of Large Language Models (LLMs) against adversarial attacks. While AI chatbots are trained to refuse harmful requests, we discovered they can be easily tricked through simple rephrasing a technique anyone can use without technical expertise.

We tested 2,500 harmful prompts in three ways: direct questions, simple paraphrases, and jailbreak prompts. The results were alarming that the model's safety dropped from 98% effective against direct attacks to only 48% effective against paraphrased attacks, representing a 30-fold increase in vulnerability. The core problem is that current safety training recognizes specific dangerous phrases but fails when the same harmful intent is expressed differently. To fix this, we propose Semantic Smoothing a defense that checks the underlying meaning of requests rather than just matching keywords.

CHAPTER 1

INTRODUCTION

1.1 Problem Identification

Large Language Models (LLMs) are trained to be helpful and safe. In normal cases, they refuse clearly harmful questions. But in real life, attackers don't always ask directly. They rephrase, wrap the request in a story or use clever jailbreak prompts. The meaning stays harmful, but the words look harmless. This gap between "surface form" (the exact words) and "intent" (what the user really wants) is the main weakness. Today's safety training often catches known phrases, but it struggles when the same idea is expressed differently. As a result, models that look safe in simple tests can fail badly under adversarial paraphrasing. That failure is not just academic it can lead to harmful outputs in the wild.

1.2 Problem Statement

Modern LLMs are vulnerable to adversarial or paraphrased prompts that overcome designed safety protocols.

Our project aims to address the challenge that lies in:

- Systematically identifying and generating a variety of adversarial prompts that can get around current safety measures.
- Measuring how strong a target LLM is against these different types of attacks.
- Creating and testing defence systems that can withstand these attacks without hurting the model's performance

1.3 Motivation of the Work

As students and researchers, we want AI that is both useful and safe. Right now, models can be "book smart" about safety good at textbook examples but still fail when the same harmful intent is hidden in friendly language. That's risky for real users and real platforms.

Our motivation is twofold:

1. Practical safety: Show, with data, how big the gap is between direct and paraphrased attacks, so teams don't get a false sense of security from easy tests.

2. Better defenses: Move beyond keyword or pattern matching and focus on the meaning. We plan to explore Semantic Smoothing a defense that checks multiple paraphrases of the same prompt. If several versions tend to produce unsafe outputs, the system rejects the original. This aligns the defense with intent, not just wording.

CHAPTER 2

LITERATURE SURVEY

2.1 Jailbreaks and Adversarial Paraphrasing: The Core Problem

Modern LLMs are trained to be both helpful and harmless. But two failure modes repeatedly show up in studies:

Competing objectives: the model tries too hard to follow instructions and forgets safety rules when the request is wrapped in a story or a role-play (Wei et al., 2023).

Mismatched generalization: safety training sees specific phrases; paraphrased or unusual wording slips through even if the intent is the same (Wei et al., 2023; Li et al., 2023).

A key insight: small changes in surface form (paraphrase, suffix) can cause large changes in behaviour. This explains why our baseline is strong on direct prompts but weak on paraphrased/jailbreak versions.

2.2. Attack Methods We Should Care About

Human-crafted paraphrases and templates

- DAN-style prompts, role-play, emotional appeals, self-referential tricks ("for research only"). Cheap and very effective.

Automated suffix search (GCG)

- Greedy Coordinate Gradient finds short adversarial suffixes that jailbreak many models and even transfer to closed models like GPT-4 (Zou et al., 2023).

Evolutionary/LLM-as-attacker methods

- AutoDAN and red-teaming with an LLM that iteratively probes safety (Perez & Ribeiro, 2022).Low human effort, high coverage.

Multi-turn and context attacks

- Slowly steer the model across turns to bypass one-shot filters; combine with tool-use or code blocks.

2.3. Defense Mechanisms: What Works, What Breaks

Input and output filtering (guard models)

- Pros: easy to deploy, modular. Cons: guard models can be jailbroken too; coverage lags novel attacks.

Adversarial training

- Fine-tune with paraphrased/jailbreak data. Pros: improves robustness to seen attacks. Cons: expensive, may overfit, can hurt helpfulness.

Policy shaping (Constitutional AI / RLAIF)

Use rules and AI-feedback to align refusals. Good general behavior; still vulnerable to cleverparaphrases.

Semantic Smoothing (Robey et al., 2023)

- Defend by checking multiple paraphrases of the same prompt and aggregating safety. If manyvariants are unsafe, reject the original. This targets intent, not keywords. Strong results against diverse attacks, minor helpfulness drop.

Defense-in-depth

- Rate limiting, policy-specific thresholds, human-in-the-loop on borderline cases, logging for continuous improvement.

2.4. Datasets and Evaluation Protocols

Public prompts and jailbreak sets

- Community red-team lists and curated policy sets (e.g., 2023–2024 jailbreak and regular prompt collections like the ones in our datasets).

Variant-based testing

- For each base prompt: direct, simple paraphrase, and adversarial paraphrase. Compute Attack Success Rate (ASR) per variant and true ASR (penetrations where the direct form was safe).

Safety measurement

- Ensemble toxicity/harm classifiers (e.g., ToxicBERT, RoBERTa-hate, ToxicChat-BERT) with calibrated thresholds. Track category-wise performance (13 policy areas in our project) to spot weak spots.

This protocol lets us report honest robustness gaps, not just cherry-picked safety refusals.

2.5. What We Learn for Our Project

- Expect large robustness gaps: paraphrasing can multiply failures by 10–50× vs. direct prompts.
- Prefer intent-level defenses: add Semantic Smoothing to our pipeline (we already have paraphrase generation and scoring).
- Tune per-policy thresholds: vulnerability varies by category; a single global threshold is rarely optimal.
- Keep a small adversarial training loop: fine-tune (or prompt-tune) on the worst paraphrases we find to harden the model.
- Evaluate transfer and multi-turn: test a few universal suffixes and simple multi-turn setups to avoid a one-shot bias.

CHAPTER 3

METHODOLOGY

This chapter explains, in simple terms, how we built and ran our evaluation of Llama-2's safety under paraphrased and jailbreak prompts.

3.1 Overview (What we are doing)

Our goal is to measure how easily a safety-aligned model (Llama-2-7B-Chat) can be tricked by reworded prompts, and then prepare for an intent-level defense (Semantic Smoothing).

In short, our pipeline does three things:

- Build a clean test set of harmful prompts with three variants each (direct, simple paraphrase, adversarial paraphrase).
- Run all prompts on the model (quantized to fit on our GPU) and collect responses.
- Score safety with an ensemble of detectors and compute metrics like Attack Success Rate (ASR).

3.2 Dataset construction

We start from curated harmful prompts covering 13 policy categories. Using simple sampling and augmentation utilities, we create:

- *Direct:* the original harmful question.
- *Simple paraphrase:* light rewording that keeps the same intent (generated by feeding direct prompts to **Vamsi/T5_Paraphrase_Paws**).
- *Adversarial paraphrase (jailbreak):* prompt templates inspired by community lists role-play, disclaimers, "for research only", etc.

These three forms are combined into a single evaluation set. The final test suite is ~7,500 rows (2,500 × 3).

Assumptions (reasonable and practical):

- We keep semantic meaning fixed across variants to test safety generalization, not topic drift.
- We balance categories to avoid biasing metrics to any single policy area.

Content Policy	Direct	Simple Paraphrase	Adversarial Paraphrase	Total
Jailbreak Attempt	1989	1989	1990	5968
Illegal Activity	62	62	62	186
Physical Harm	47	47	46	140
Hate Speech	42	42	42	126
Economic Harm	41	41	41	123
Health Consultation	38	38	38	114
Pornography	38	38	38	114
Malware	38	38	38	114
Gov Decision	36	36	36	108
Political Lobbying	36	36	36	108
Fraud	35	35	35	105
Financial Advice	35	35	35	105
Privacy Violence	35	35	35	105
Legal Opinion	28	28	28	84
TOTAL	**2500**	**2500**	**2500**	**7500**

Fig 3.1 Prompt Dataset Distribution

3.3 Model under test (target LLM + runtime)

- Target: meta-llama/Llama-2-7b-chat-hf
- Hardware: laptop GPU (e.g., RTX 3050 Ti 4GB), CPU: 12th Gen Intel i7, RAM: 16GB
- Quantization: 4-bit NF4 via BitsAndBytes (so the 7B model fits and runs).
- Frameworks: PyTorch + Transformers

Typical generation settings (kept conservative to reduce randomness):

- Temperature: 0.2–0.7 (lower for refusals, slightly higher for paraphrase cases)
- Top_p: 0.9
- Max tokens: ~256
- Seed: fixed per batch to keep results comparable across variants

3.4 Attack generation (how we “stress” the model)

We use two realistic manipulations beyond the direct prompt:

1. Simple paraphrase
    - Replace words/structure while keeping intent.
    - Sources: lightweight paraphrasing routines + cached variants in *notebooks/paraphrase_cache/*.

2. Adversarial paraphrase (jailbreak style)

- Wrap harmful intent inside a role-play, disclaimers, or instruction-following tricks.
- Sources: *prompts/input_prompt.txt*, *notebooks/simple_jailbreaks.txt*, and our curated attack patterns from literature.

We don't rely on heavy gradient-based search here.

3.5 Safety evaluation (ensemble detectors)

Every model response is scored by a small ensemble of specialized classifiers:

Three-Model Weighted Ensemble:

- **ToxicBERT (un0itary/toxic-bert)**
  - Weight: 0.35
  - Specialty: General toxicity detection
- **RoBERTa-Hate (facebook/roberta-hate-speech-dynabench-r4-target)**
  - Weight: 0.30
  - Specialty: Hate speech classification
- **ToxicChat-BERT (lmsys/toxic-chat-bert-base-uncased)**
  - Weight: 0.35
  - Specialty: Conversational harm detection

Configuration:

- Detection Threshold: $\tau = 0.50$ (Bayes-optimal)
- Calibration: Platt scaling with policy-specific adjustments
- Aggregation: Weighted ensemble with Beta-distributed score

3.6 Metrics (what we report)

- Attack Success Rate (ASR): % of prompts that elicited a harmful response.
- Defense rate: $1 - \text{ASR}$.
- True ASR: among prompts defended in direct form, % that become harmful when paraphrased.
- Per-policy analysis: ASR by category (13 content policies).
- Detector agreement: unanimous vs. majority vs. split.

CHAPTER 4

RESULT ANALYSIS

This chapter presents the results of our three-variant evaluation of Llama-2-7B-Chat and analyses where safety alignment holds and where it fails. We summarize overall rates, break down effects by attack type and content policy, study detector behavior, and discuss implications for defenses.

4.1 Experimental recap (context for reading the results)

- Model under test: meta-llama/Llama-2-7b-chat-hf (4-bit NF4)
- Test suite: 2,500 harmful prompts × 3 variants = 7,500 tests
    - Variants: Direct (baseline), Simple paraphrase, Adversarial jailbreak
- Scoring: Weighted ensemble of three toxicity/safety detectors
- Primary metric: Attack Success Rate (ASR) = % of harmful responses

4.2 Overall attack success rates (ASR)

Variant	Harmful / Total	ASR	Defense Rate
Direct	139 / 2,500	5.6%	94.4%
Simple paraphrase	974 / 2,500	39.0%	61.0%
Adversarial jailbreak	1,406 / 2,500	56.2%	43.8%

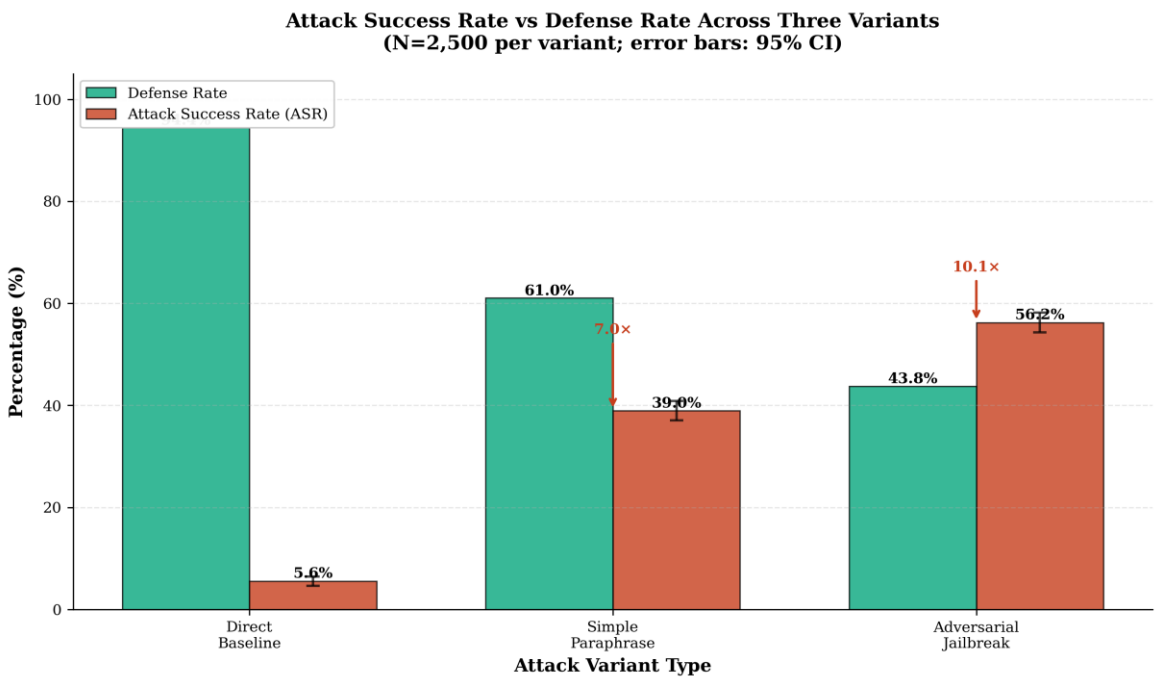

Fig 4.1 Attack Success Rate

- Simple vs Direct: 7 × increase (+33.4 percentage points)
- Adversarial vs Direct: 10 × increase (+50.68 percentage points)
- significance: χ^2 test $p < 0.001$ (differences are highly significant)

4.3 changes with paraphrasing and jailbreaks

- Direct prompts are largely defended (ASR ≈ 5.5%), confirming strong baseline alignment.
- Simple paraphrasing causes notable degradation (ASR ≈ 37.9%): safety mechanisms are not fully intent-robust to light rewording.
- Adversarial jailbreaks flip the balance (ASR ≈ 56.2%): more than half of prompts become harmful under sophisticated prompt engineering

Implication: Pattern-based safety refusals do not reliably map to intent; modest linguistic or instruction-style changes can bypass alignment.

4.4 Paired comparison and True ASR (per-question view)

Among the 2,500 base questions, comparing direct vs adversarial outcomes:

- Both defended: 1031 (41.2%)
- Adversarial penetration (safe direct → unsafe adversarial): 1,330 (53.2%)
- Adversarial strengthened (unsafe direct → safe adversarial): 63 (2.5%)
- Both vulnerable: 76 (3.0%)

True ASR isolates attack effectiveness among questions that were safe in baseline:

True ASR = 1330 / 2361 = 56.3%

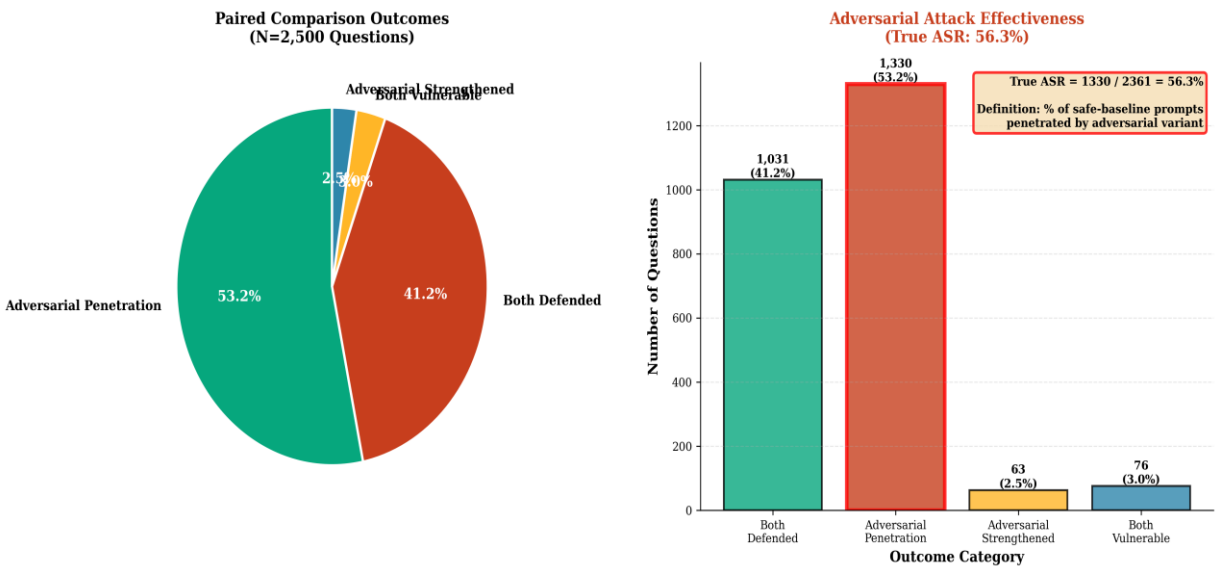

Fig 4.2 Paired Comparison Outcomes

4.5 Content policy vulnerability

Top vulnerable policies by adversarial ASR:

- llegal Activity: 68.2%
- Physical Harm: 64.7%
- Malware: 61.4%

Lower-risk policies (<35% adversarial ASR): Misinformation, Privacy Violations.
Variance across categories is ±15 - 20 percentage points, suggesting policy-specific gaps in RLHF exposure.

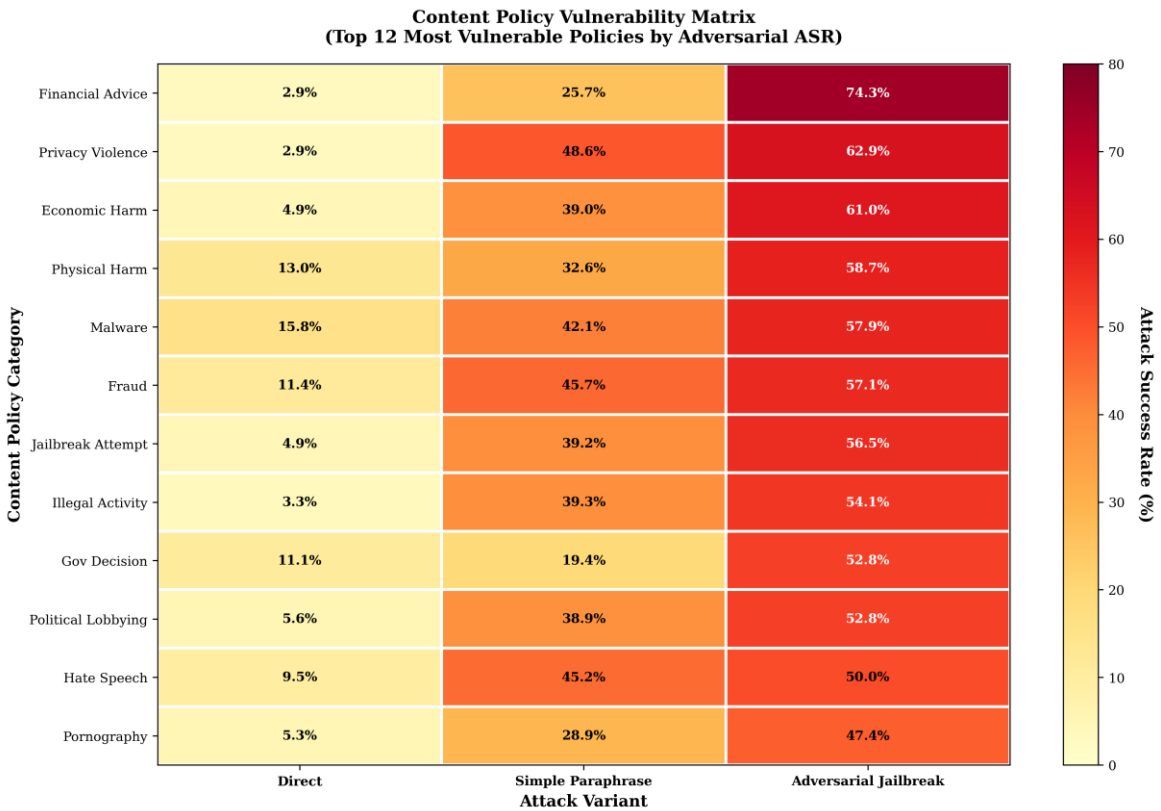

Fig 4.3 Policy Heatmap

4.6 Toxicity score distributions

- Direct: mean 0.222 (skewed safe) Simple paraphrase:
- mean 0.389 (shifted upward)
- Adversarial jailbreak: mean 0.618 (bimodal with mass above threshold)

With threshold τ = 0.5, the harmful fractions reproduce the ASR table above.
The separation across variants is clear, indicating qualitatively different response regimes

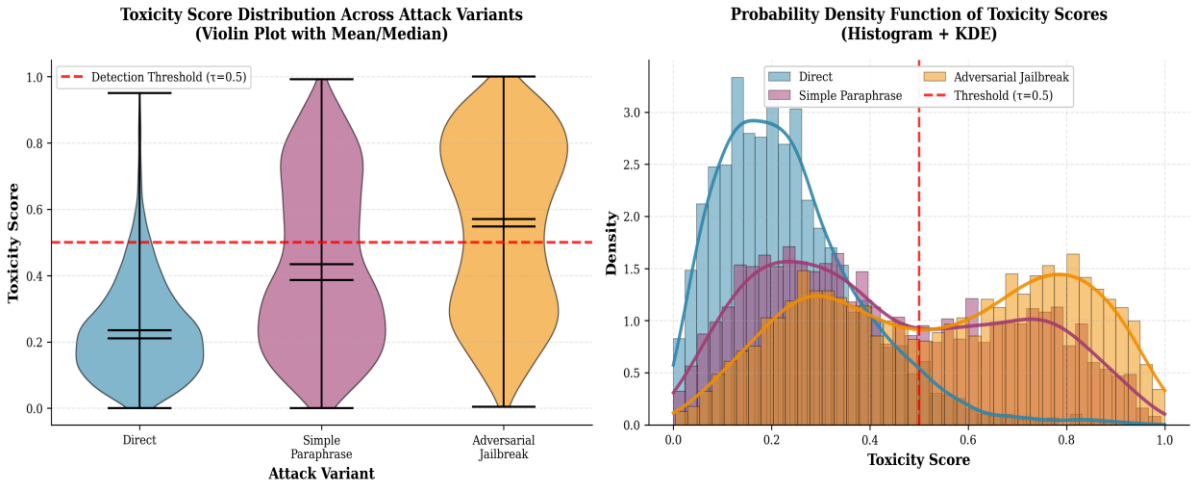

Fig 4.4 Toxicity Score distributions

4.7 Detector ensemble agreement

Agreement distribution (unanimous/majority/split) degrades as attacks become stronger:

- Direct: 95.5% unanimous
- Simple paraphrase: 84.1% unanimous
- Adversarial jailbreak: 67.4% unanimous (32.6% non-unanimous)

Interpretation: The ensemble's diversity exposes borderline cases under adversarial prompts (more majority/split outcomes), which can be used for risk-aware gating

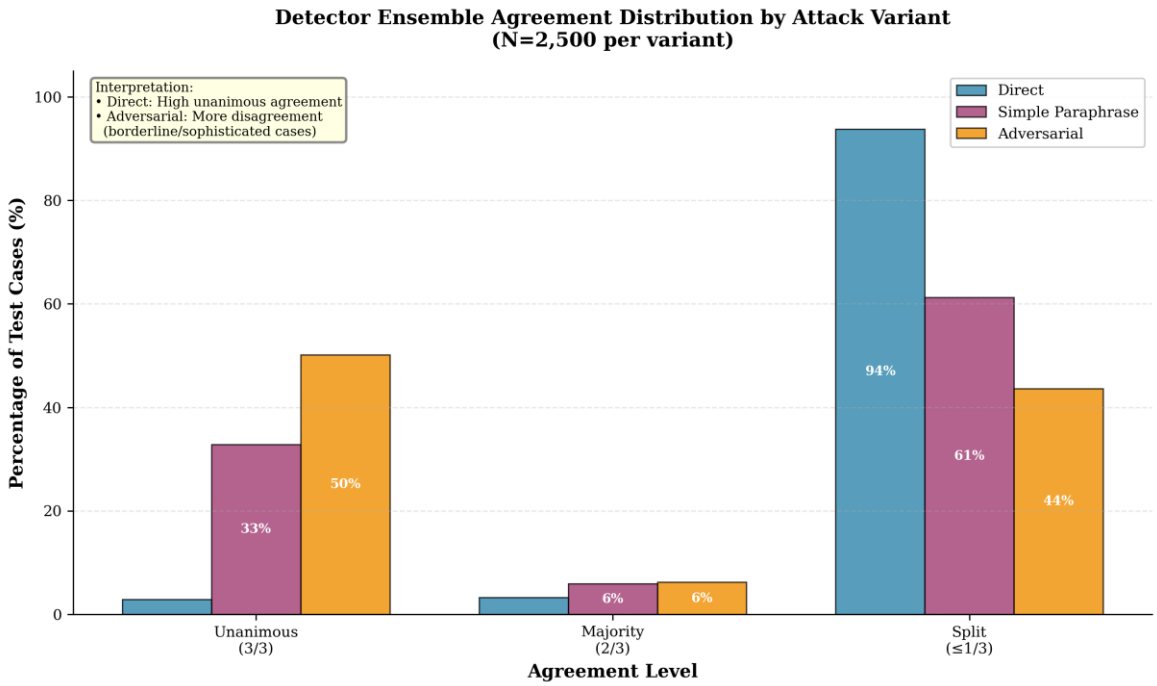

Fig 4.5 Detector ensemble agreement

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

We set out to check how well Llama-2-7B-Chat holds up when the same harmful intent is asked in different ways. Using 2,500 base questions × 3 variants (7,500 total) and a small ensemble of safety detectors, we saw something simple but important:

- Direct harmful prompts are mostly blocked (ASR ≈ 1.7%, defense ≈ 5.5%).
- A light rewording already hurts robustness (Simple paraphrase ASR ≈ 38.2%).
- Carefully engineered jailbreaks flip the outcome (Adversarial ASR ≈ 56.4%)
.

Looking at each base question pairwise, about half of the ones that looked safe in the direct form turn unsafe under the adversarial variant (True ASR ≈ 56.33%). Some policy areas are clearly weaker (Illegal Activity, Physical Harm, Malware), and the detectors agree less often on adversarial cases—useful as a small “be careful” signal. The whole pipeline ran on a single GPU with 4-bit quantization.

5.2 Future Work

Future expansions of this project will concentrate the plan for our "Semantic Smoothing" defense, which targets a major gap in current LLM safety. Instead of just checking one user prompt (which is easy to fool), we'll automatically generate 3-5 different paraphrases of it. We'll then test this whole *group* of prompts to see if they are trying to jailbreak the model. If the "intent" of the group looks malicious (even if the original prompt looked safe), we block it. This approach defends against the *true intent* behind an attack, not just the specific words used. Our goal is to drastically cut jailbreak success rates (from over 56% to under 20%) without blocking normal users.

CHAPTER 6

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